

Electronics Workshop II, Project-2

Continuous Time Analog Calculator

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Abstract—Modern edge-device applications increasingly demand localized, on-chip processing with stringent constraints on power, memory, and latency. Traditional digital architectures struggle to compute complex non-linear functions, such as logarithms, efficiently. Digital implementations inherently require continuous sampling (ADCs), quantization, and the execution of algorithmic approximations—such as Taylor series expansions or memory-heavy Lookup Tables (LUTs)—consuming significant clock cycles and silicon area. To address these bottlenecks, this project presents a Continuous-Time Analog Calculator. By exploiting the fundamental device physics of discrete PN junctions, operational amplifiers, and optimized resistor networks, this architecture bypasses algorithmic computation entirely. The system performs instantaneous, zero-latency physical computation at the speed of electron flow, completely eliminating the need for clock-driven execution and digital memory storage.

Index Terms—Continuous-Time Analog, Operational Amplifiers, Shockley Diode Equation, Edge Computing, Low Power Devices.

I. INTRODUCTION

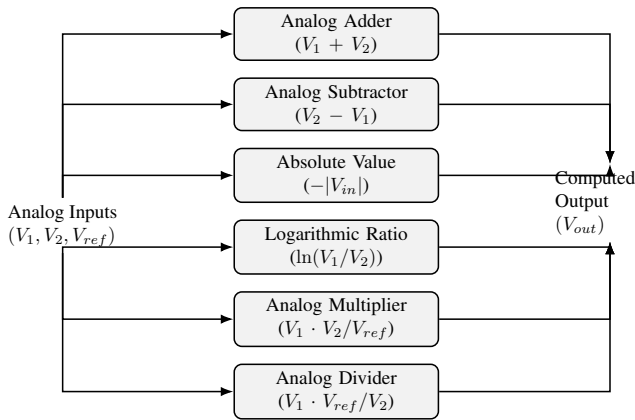


Fig. 1. High-Level System Block Diagram of the Continuous-Time Analog ALU Architecture.

The objective of this project is to process analog input voltages and execute mathematical operations, yielding an output voltage that directly corresponds to the computed result. To construct these circuits from the ground up, we harness the fundamental mathematical and physical properties of op-

erational amplifiers (op-amps), discrete diodes, and precision resistors.

While a combination of op-amps and resistors is sufficient for linear voltage manipulation (such as addition and subtraction), operations like multiplication, division, and logarithms are inherently non-linear. Therefore, purely resistive feedback networks are inadequate. To achieve these advanced functions, we integrate discrete diodes into the negative feedback loop of the op-amp. This architecture actively exploits the physics of the PN junction to convert linear voltage inputs into logarithmic or exponential scales, forming the core computational engine of our continuous-time calculator.

Project Goal: To design, simulate, and physically demonstrate that through rigorous component selection, impedance optimization, and precise calibration, it is entirely possible to achieve sub-4% error margins in discrete analog computation.

Core Components:

- **LM358P Operational Amplifier:** This op-amp serves as the primary active component for all our mathematical operations. It was selected because it provides highly reliable performance with a relatively low offset voltage ($\sim 2\text{-}3\text{mV}$) at a highly cost-effective price point. While it is not a dedicated precision op-amp (which are significantly more expensive and specifically manufactured to eliminate offset issues), utilizing the LM358P proves that exceptional accuracy can still be achieved. Through careful circuit calibration and subspace mapping, we successfully harness the op-amp's inherent properties while mitigating its hardware limitations.
- **1N4148 Fast Switching Diodes:** These discrete diodes execute the core non-linear math by adhering strictly to the Shockley diode equation in the forward-biased region:

$$I_D = I_S e^{\frac{V_D}{\eta V_T}} \quad (1)$$

The 1N4148 was chosen because it maintains this precise logarithmic voltage-current relationship even at high frequencies. Consequently, the diode will never act as an error-inducing bottleneck for high-frequency signals; the maximum operating frequency of the calculator will

always be cleanly bounded by the op-amp's slew rate. Although our current prototype is optimized for DC computation, this robust component choice ensures the underlying architecture is fully capable of processing dynamic, continuous-time AC signals.

- **Precision Resistors:** The implemented circuits utilize a wide array of resistor values, spanning from hundreds of ohms to hundreds of kilo-ohms. These specific impedance networks were not chosen arbitrarily; they were carefully scaled to balance op-amp current-drive limits against input bias current errors, which will be discussed in detail throughout the architectural breakdown of each computational block.

II. DESIGN

In this section, we detail the architecture and hardware implementation of the core computational blocks that constitute the continuous-time analog calculator. The circuits we are going to design and calibrate are the following:

- **Analog Adder**
- **Analog Subtractor**
- **Absolute Value Circuit**
- **Logarithmic Amplifier**
- **Analog Multiplier**
- **Analog Divider**

A. Analog Adder

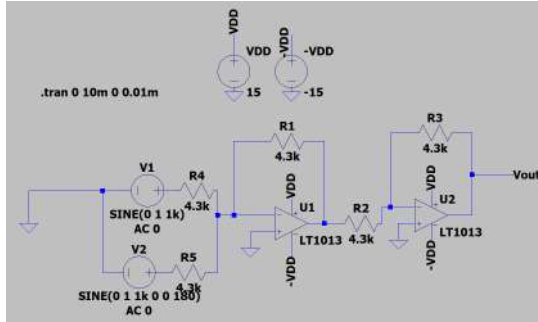


Fig. 2. Analog Addition Circuit

The adder consists of two stages: an inverting summer (U1) and a unity-gain inverter (U2). Let all identically valued resistors (4.3 kΩ) be denoted as R .

Applying the virtual ground concept at the inverting terminal of U1 ($V_{x1} = 0$ V) and Kirchhoff's Current Law (KCL):

$$\frac{V_1 - 0}{R} + \frac{V_2 - 0}{R} + \frac{V_{mid} - 0}{R} = 0 \quad (2)$$

Solving for the intermediate voltage V_{mid} :

$$V_{mid} = -(V_1 + V_2) \quad (3)$$

For the second stage (U2), applying KCL at its virtual ground node ($V_{x2} = 0$ V):

$$\frac{V_{mid} - 0}{R} + \frac{V_{out} - 0}{R} = 0 \quad (4)$$

Solving for V_{out} :

$$V_{out} = -V_{mid} \quad (5)$$

Substituting Equation (2) into Equation (4) yields the final non-inverted output:

$$V_{out} = -[-(V_1 + V_2)] = V_1 + V_2 \quad (6)$$

B. Analog Subtractor

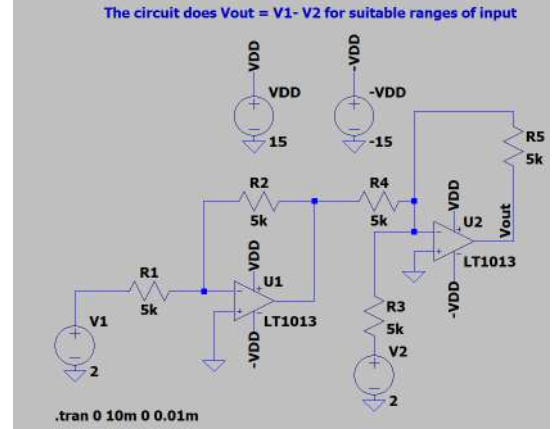


Fig. 3. Analog Subtraction Circuit

The analog subtractor (difference amplifier) computes the difference between two inputs, V_1 and V_2 , using a single operational amplifier. Assuming a standard four-resistor network where all resistors are equal (4.3 kΩ), we denote them simply as R . Let V_2 be applied to the non-inverting path and V_1 to the inverting path.

First, we determine the voltage at the non-inverting terminal (V_+). Since no current flows into the ideal op-amp, the resistors form a simple voltage divider from V_2 to ground:

$$V_+ = V_2 \left(\frac{R}{R + R} \right) = \frac{V_2}{2} \quad (7)$$

Next, apply Kirchhoff's Current Law (KCL) at the inverting terminal (V_-). The sum of currents leaving the node through the input resistor and the feedback resistor is zero:

$$\frac{V_- - V_1}{R} + \frac{V_- - V_{out}}{R} = 0 \quad (8)$$

Multiplying the entire equation by R yields:

$$V_- - V_1 + V_- - V_{out} = 0 \quad (9)$$

$$2V_- - V_1 = V_{out} \quad (10)$$

Due to the negative feedback loop, the virtual short concept applies, forcing the voltages at the inverting and non-inverting terminals to be equal ($V_- = V_+$). Substituting Equation (1) into Equation (4):

$$2 \left(\frac{V_2}{2} \right) - V_1 = V_{out} \quad (11)$$

Simplifying this gives the final transfer function for the analog subtractor:

$$V_{out} = V_2 - V_1 \quad (12)$$

C. Absolute Value Circuit

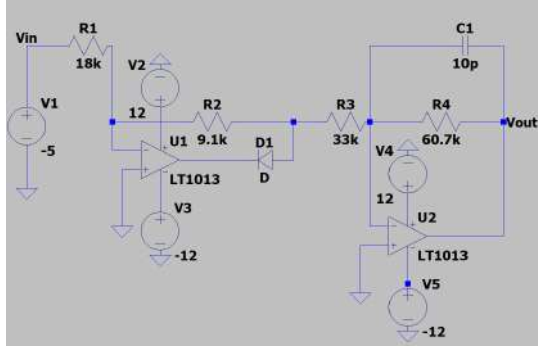


Fig. 4. Absoulte Value Cricuit

This circuit functions as an inverting precision rectifier, effectively computing $V_{out} \approx -|V_{in}|$. The architecture relies on the state of diode D1 to dynamically alter the signal path based on the polarity of the input voltage.

1) *Case 1: Positive Input Voltage ($V_{in} > 0$):* When V_{in} is positive, the operational amplifier U1 attempts to sink current, driving its output negative to maintain virtual ground. This negative voltage reverse-biases diode D1, turning it OFF and breaking the feedback loop of U1.

With U1 effectively removed from the signal path, the current flows directly from the input through the series combination of R_1 , R_2 , and R_3 into the virtual ground of U2. This creates a single inverting amplifier stage with a large input resistance. Applying Kirchhoff's Current Law at the inverting terminal of U2:

$$\frac{V_{in} - 0}{R_1 + R_2 + R_3} + \frac{V_{out} - 0}{R_4} = 0 \quad (13)$$

Solving for V_{out} :

$$V_{out} = -V_{in} \left(\frac{R_4}{R_1 + R_2 + R_3} \right) \quad (14)$$

Substituting the component values from the schematic:

$$V_{out} = -V_{in} \left(\frac{60.7 \text{ k}\Omega}{18 \text{ k}\Omega + 9.1 \text{ k}\Omega + 33 \text{ k}\Omega} \right) = -V_{in} \left(\frac{60.7 \text{ k}\Omega}{60.1 \text{ k}\Omega} \right) \approx -1.01V_{in} \quad (15)$$

Because the diode is off, the circuit bypasses the first active stage, yielding roughly unity gain and a negative output.

2) *Case 2: Negative Input Voltage ($V_{in} < 0$):* When V_{in} is negative, U1 must source current to maintain its virtual ground, driving its output positive. This forward-biases diode D1, turning it ON. The circuit now operates as a cascade of two distinct op-amp gain stages.

For the first stage (U1), the intermediate voltage at the node between R_2 and R_3 , denoted as V_{mid} , is calculated using the standard inverting amplifier formula:

$$V_{mid} = -V_{in} \left(\frac{R_2}{R_1} \right) = -V_{in} \left(\frac{9.1 \text{ k}\Omega}{18 \text{ k}\Omega} \right) \approx -0.505V_{in} \quad (16)$$

This intermediate voltage is then processed by the second inverting stage (U2):

$$V_{out} = -V_{mid} \left(\frac{R_4}{R_3} \right) = -V_{mid} \left(\frac{60.7 \text{ k}\Omega}{33 \text{ k}\Omega} \right) \approx -1.839V_{mid} \quad (17)$$

Substituting V_{mid} from Equation (4) into Equation (5) yields the total transfer function:

$$V_{out} = -(-0.505V_{in}) \times 1.839 \approx 0.929V_{in} \quad (18)$$

Since V_{in} is negative, the resulting V_{out} remains negative. Because the magnitude multiplier is 0.929 (which is less than 1.0), the cascaded stage architecture naturally results in a slightly attenuated output value compared to the true absolute magnitude.

D. Logarithm Operation

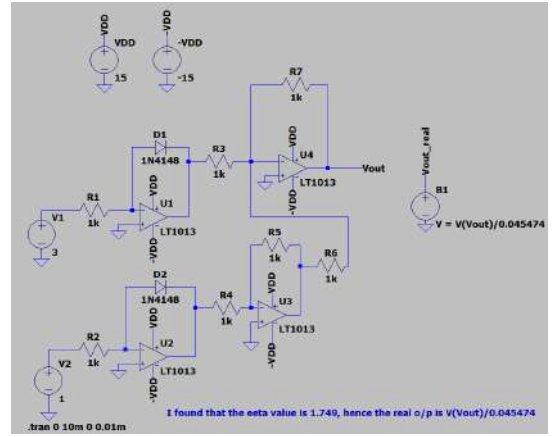


Fig. 5. Logarithm Calculator

This circuit architecture utilizes two parallel logarithmic amplifiers followed by an analog subtraction stage. By exploiting the logarithmic properties of the Shockley diode equation, the circuit inherently converts subtraction into division, yielding the natural logarithm of the ratio of the two inputs. Let all identical $1 \text{ k}\Omega$ resistors be denoted as R .

Stage 1: Parallel Logarithmic Amplifiers (U1 and U2): For the first op-amp (U1), the input voltage V_1 drives a current through R_1 . Because the inverting terminal is at virtual ground (0 V) and ideal op-amps draw no input current, this entire current flows through the feedback diode D1.

$$I_{D1} = \frac{V_1}{R} \quad (19)$$

The voltage across the diode, V_{D1} , is the difference between the virtual ground and the output voltage of U1 (V_{log1}):

$$V_{D1} = 0 - V_{log1} = -V_{log1} \quad (20)$$

Substituting this into the forward-biased Shockley diode equation:

$$I_{D1} \approx I_s e^{\frac{V_{D1}}{\eta V_T}} \implies \frac{V_1}{R} = I_s e^{\frac{-V_{log1}}{\eta V_T}} \quad (21)$$

Solving for the intermediate output V_{log1} :

$$V_{log1} = -\eta V_T \ln \left(\frac{V_1}{RI_s} \right) \quad (22)$$

By symmetry, the second logarithmic amplifier (U2) processes V_2 identically:

$$V_{log2} = -\eta V_T \ln \left(\frac{V_2}{RI_s} \right) \quad (23)$$

2) *Stage 2: Inversion and Summation (U3 and U4)*: To subtract the logarithmic values, the output of U2 (V_{log2}) is first passed through a unity-gain inverting amplifier (U3), since $R_4 = R_5 = R$. The output of U3 is simply $-V_{log2}$.

The final stage is an inverting summing amplifier (U4). Applying Kirchhoff's Current Law at its virtual ground:

$$\frac{V_{log1} - 0}{R_3} + \frac{(-V_{log2}) - 0}{R_6} + \frac{V_{out} - 0}{R_7} = 0 \quad (24)$$

Since $R_3 = R_6 = R_7 = R$, the equation simplifies to:

$$V_{out} = -(V_{log1} - V_{log2}) = V_{log2} - V_{log1} \quad (25)$$

3) *Final Transfer Function*: Substituting Equations (4) and (5) into the summation equation:

$$V_{out} = \left[-\eta V_T \ln \left(\frac{V_2}{RI_s} \right) \right] - \left[-\eta V_T \ln \left(\frac{V_1}{RI_s} \right) \right] \quad (26)$$

Rearranging the terms:

$$V_{out} = \eta V_T \ln \left(\frac{V_1}{RI_s} \right) - \eta V_T \ln \left(\frac{V_2}{RI_s} \right) \quad (27)$$

Factoring out the constants:

$$V_{out} = \eta V_T \left[\ln \left(\frac{V_1}{RI_s} \right) - \ln \left(\frac{V_2}{RI_s} \right) \right] \quad (28)$$

Applying logarithmic quotient rules ($\ln(a) - \ln(b) = \ln(a/b)$) mathematically cancels out both the reverse saturation current (I_s) and the resistance (R):

$$V_{out} = \eta V_T \ln \left(\frac{V_1}{\frac{V_2}{V_r}} \right) = \eta V_T \ln \left(\frac{V_1}{V_2} \right) \quad (29)$$

4) *Thermodynamic Calibration and Scaling Factor*: While the analog circuit executes the logarithmic math flawlessly via the physics of the PN junction, the raw output voltage is inherently bounded by thermodynamics. The mathematical result is multiplied by the thermal voltage ($V_T \approx 25.85$ mV at room temperature) and the specific diode's ideality factor ($\eta \approx 1.749$).

This physical reality compresses the pure mathematical computation into a narrow millivolt-scale window ($\eta V_T \approx 0.04547$ V). Therefore, to extract the pure, unitless mathematical result for downstream utilization, a deliberate calibration step must be introduced. By characterizing the diode to find η and measuring ambient temperature to define V_T , we normalize the output by applying a scaling scalar $K = \frac{1}{\eta V_T}$. Dividing the raw output voltage by this hardware-derived constant (0.04547) strips away the physical thermodynamic constraints, yielding the absolute, high-precision logarithmic ratio.

E. Analog Multiplier

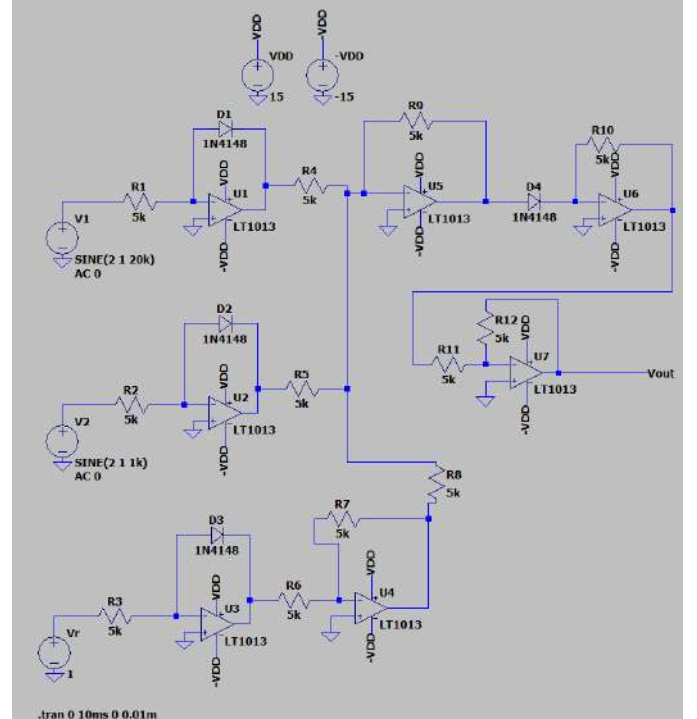


Fig. 6. Analog Multiplier

The analog multiplier (Figure 1) computes the product of two input voltages, V_1 and V_2 , scaled by a reference voltage, V_r . It operates by taking the logarithms of the inputs, summing them, and passing the result through an antilogarithmic amplifier. All resistors are identically sized at $5 \text{ k}\Omega$ and will be denoted as R .

1) *Stage 1: Logarithmic Amplifiers*: The first stage consists of three parallel logarithmic amplifiers (U1, U2, U3) processing V_1 , V_2 , and V_r respectively. Based on the forward-biased Shockley diode equation, their outputs are:

$$V_{log1} = -\eta V_T \ln \left(\frac{V_1}{RI_s} \right) \quad (30)$$

$$V_{log2} = -\eta V_T \ln \left(\frac{V_2}{RI_s} \right) \quad (31)$$

$$V_{logr} = -\eta V_T \ln \left(\frac{V_r}{RI_s} \right) \quad (32)$$

2) *Stage 2: Inversion and Summation*: To perform division by V_r , the signal V_{logr} is passed through a unity-gain inverting amplifier (U4), yielding $-V_{logr}$.

Next, V_{log1} , V_{log2} , and the inverted reference $-V_{logr}$ are fed into an inverting summing amplifier (U5). The output of the summer, V_{sum} , is:

$$V_{sum} = -(V_{log1} + V_{log2} - V_{logr}) \quad (33)$$

Substituting the logarithmic equations and factoring out ηV_T :

$$V_{sum} = \eta V_T \left[\ln \left(\frac{V_1}{RI_s} \right) + \ln \left(\frac{V_2}{RI_s} \right) - \ln \left(\frac{V_r}{RI_s} \right) \right] \quad (34)$$

Applying the properties of logarithms ($\ln(a) + \ln(b) - \ln(c) = \ln(ab/c)$):

$$V_{sum} = \eta V_T \ln \left(\frac{\frac{V_1}{RI_s} \cdot \frac{V_2}{RI_s}}{\frac{V_r}{RI_s}} \right) = \eta V_T \ln \left(\frac{V_1 V_2}{V_r RI_s} \right) \quad (35)$$

3) *Stage 3: Antilogarithmic Amplifier:* The intermediate voltage V_{sum} is applied across diode D4 at the input of U6. The current through D4 is exponentially related to V_{sum} :

$$I_{D4} = I_s e^{\frac{V_{sum}}{\eta V_T}} \quad (36)$$

Because U6 is an inverting transimpedance configuration, its output is $V_{u6} = -I_{D4}R$. To correct the polarity, a final unity-gain inverting amplifier (U7) is used, giving the final output:

$$V_{out} = -V_{u6} = I_{D4}R = RI_s e^{\frac{V_{sum}}{\eta V_T}} \quad (37)$$

Substituting V_{sum} from Equation (6) into the exponential function completely cancels the logarithmic operation and the ηV_T terms:

$$V_{out} = RI_s \left(\frac{V_1 V_2}{V_r RI_s} \right) \quad (38)$$

The reverse saturation current (I_s) and resistance (R) neatly cancel out, leaving the final transfer function:

$$V_{out} = \frac{V_1 \cdot V_2}{V_r} \quad (39)$$

F. Analog Divider

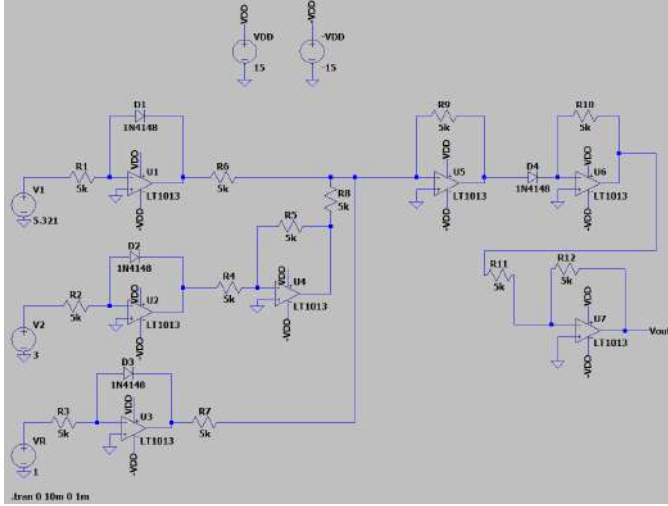


Fig. 7. Analog Divider

The analog divider (Figure 2) utilizes the exact same core architecture as the multiplier, but alters the routing to the summation stage to compute V_1 divided by V_2 , scaled by a reference voltage V_r .

1) *Routing and Summation:* The initial logarithmic outputs (V_{log1} , V_{log2} , V_{logr}) remain identical to Equations 1 through 3. However, in this configuration, the V_{log2} signal is passed through the unity-gain inverter (U4), yielding $-V_{log2}$.

The inverting summing amplifier (U5) now adds V_{log1} , V_{logr} , and the inverted $-V_{log2}$:

$$V_{sum} = -(V_{log1} + V_{logr} - V_{log2}) \quad (40)$$

Substituting the logarithmic equations:

$$V_{sum} = \eta V_T \left[\ln \left(\frac{V_1}{RI_s} \right) + \ln \left(\frac{V_r}{RI_s} \right) - \ln \left(\frac{V_2}{RI_s} \right) \right] \quad (41)$$

Applying logarithmic properties to combine the terms:

$$V_{sum} = \eta V_T \ln \left(\frac{\frac{V_1}{RI_s} \cdot \frac{V_r}{RI_s}}{\frac{V_2}{RI_s}} \right) = \eta V_T \ln \left(\frac{V_1 V_r}{V_2 RI_s} \right) \quad (42)$$

2) *Final Transfer Function:* Passing this V_{sum} through the identical antilogarithmic and inversion stages (U6 and U7) exponentiates the result:

$$V_{out} = RI_s e^{\frac{V_{sum}}{\eta V_T}} \quad (43)$$

Substituting Equation (12) into Equation (13):

$$V_{out} = RI_s \left(\frac{V_1 V_r}{V_2 RI_s} \right) \quad (44)$$

Canceling the device-specific parameters (I_s and R) yields the final division transfer function:

$$V_{out} = \frac{V_1 \cdot V_r}{V_2} \quad (45)$$

Note: The reference voltage V_r acts as a gain-scaling factor to ensure the final output remains within the voltage rails of the operational amplifiers.

G. Practical Resistor Sizing Limitations

While ideal circuit theory suggests that only resistor *ratios* matter, physical operational amplifiers dictate strict bounds on the absolute resistance values used in our circuits.

- **The Lower Bound (Current Limits):** Using excessively small resistors forces the op-amp to drive dangerously high currents. This causes the output to clip and, as we observed during early prototyping, can lead to catastrophic thermal failure (destroying the IC).
- **The Upper Bound (Bias Current Errors):** Real op-amps are not ideal and draw a small, continuous DC input bias current (I_B). If the feedback resistors are too large, this current creates a significant, unintended voltage drop ($V_{error} \approx I_B \times R$) that corrupts the analog math.

Therefore, the specific values used in our final designs (e.g., 4.3 k Ω , 5 k Ω , 33 k Ω) represent an empirically tested "sweet spot." They are large enough to protect the op-amp from thermal overload, yet small enough to render input bias current errors negligible.

III. LTSPICE SIMULATIONS

In this section, we present the simulation results for each of the core computational blocks[cite: 1]. These simulations were performed using LTSpice to validate the theoretical derivations and optimize component values before physical hardware implementation[cite: 1].

A. Analog Adder Simulation

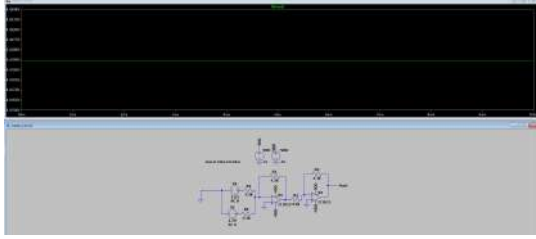


Fig. 8. Adding 2.63 and 1.79

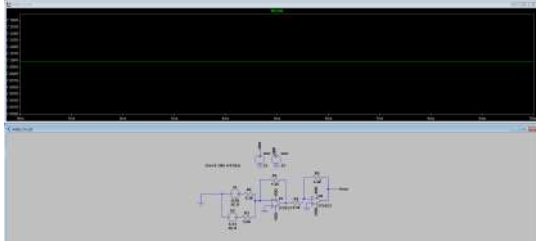


Fig. 9. Adding 0.76 and 6.34

TABLE I
ADDITION CIRCUIT EXPERIMENTAL RESULTS

V_1 (V)	V_2 (V)	V_{out} (V)	V_{exp} (V)	% Error
2.63	1.79	4.4199	4.42	0.002%
0.76	6.34	7.10	7.10	0.000%
5.43	2.09	7.520	7.52	0.000%
0.23	1.1	1.32	1.33	0.752%
8.21	1.16	9.3695	9.37	0.005%
Average Error Percentage				0.152%

We see how precise LTSpice circuit actually is, but when we add smaller and smaller values of voltages, we see a decent error since the current is now comparable to the leakage current of the opamp which invokes non-idealities

B. Analog Subtractor Simulation

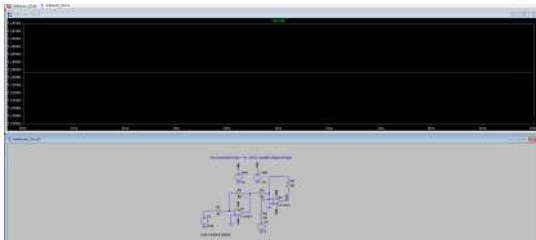


Fig. 10. subtract 3.24 and 2

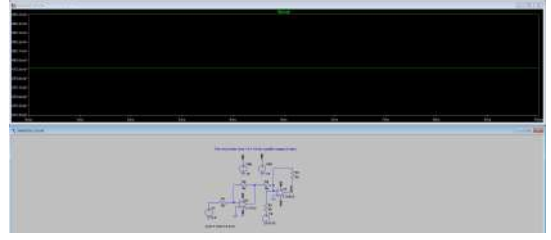


Fig. 11. Subtract 5.2 and 4.72

TABLE II
SUBTRACTION CIRCUIT EXPERIMENTAL RESULTS

V_1 (V)	V_2 (V)	V_{out} (V)	V_{exp} (V)	% Error
3.24	2.00	1.24	1.24	0.000%
5.20	4.72	0.4799	0.48	0.021%
3.00	3.00	39 μ	0.00	—
9.07	0.56	8.51	8.51	0.000%
3.50	2.16	1.339	1.34	0.075%
Average Error Percentage				0.024%

C. Absolute Value Circuit Simulation

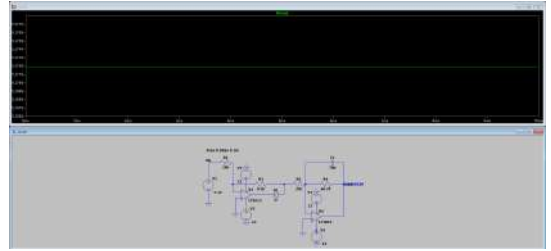


Fig. 12. absolute value of -5.32

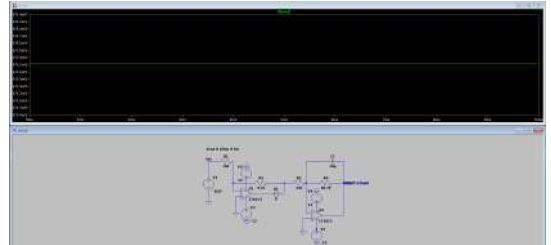


Fig. 13. Absolute Value of -0.67

TABLE III
ABSOLUTE VALUE CIRCUIT EXPERIMENTAL RESULTS

V_1 (V)	V_{out} (V)	V_{exp} (V)	% Error
-5.32	5.372	5.32	0.977%
-0.67	0.6757	0.67	0.851%
-1.69	1.7058	1.69	0.935%
0	534 μ	0.00	—
-8.54	8.624	8.54	0.984%
Average Error Percentage			0.937%

D. Logarithmic Ratio Calculator Simulation

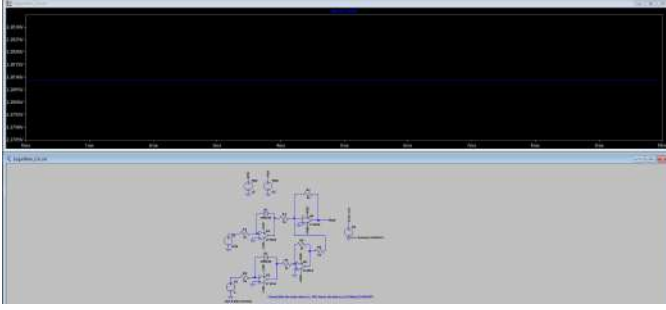


Fig. 14. Natural log of 8.93

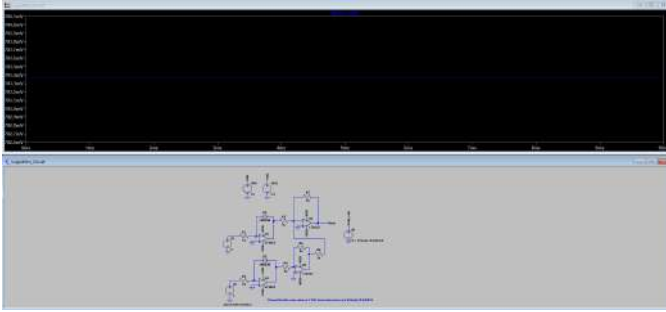


Fig. 15. Natural Log of 2

TABLE IV
LOGARITHM CIRCUIT EXPERIMENTAL RESULTS

V_1 (V)	V_{out} (V)	V_{exp} (V)	% Error
1.00	187 μ	0.0000	—
2.00	0.7030	0.6931	1.428%
3.57	1.3004	1.2726	2.185%
0.64	-0.4490	-0.4463	0.605%
8.93	2.2809	2.1894	4.179%
Average Error Percentage			2.099%

E. Analog Multiplier Simulation

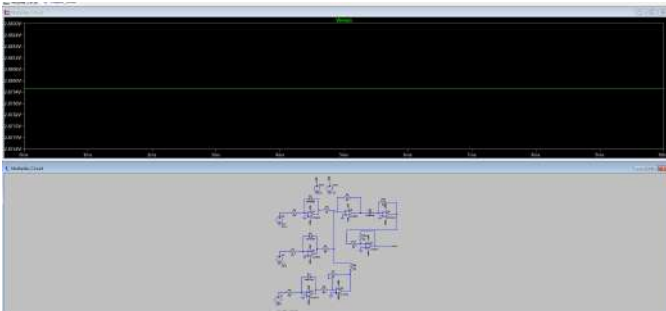


Fig. 16. multiplying 0.9 and 3.2

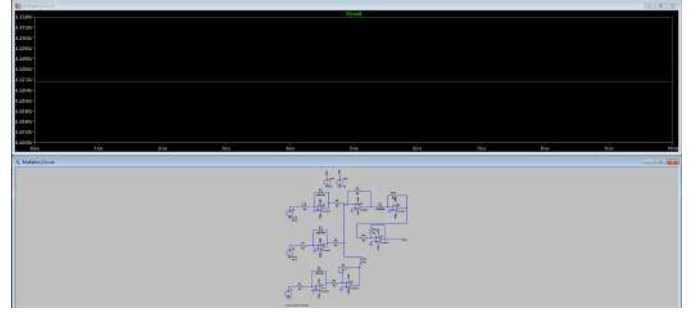


Fig. 17. multiplying 2.54 and 1.67

TABLE V
MULTIPLICATION CIRCUIT EXPERIMENTAL RESULTS

V_1 (V)	V_2 (V)	V_{mul} (V)	V_{exp} (V)	% Error
0.90	3.20	2.8790	2.8800	0.035%
2.54	1.67	4.2227	4.2418	0.450%
4.72	1.88	8.7880	8.8736	0.965%
0.43	0.67	0.2879	0.2881	0.069%
7.30	0.20	1.4781	1.4600	1.240%
Average Error Percentage				0.552%

F. Analog Divider Simulation

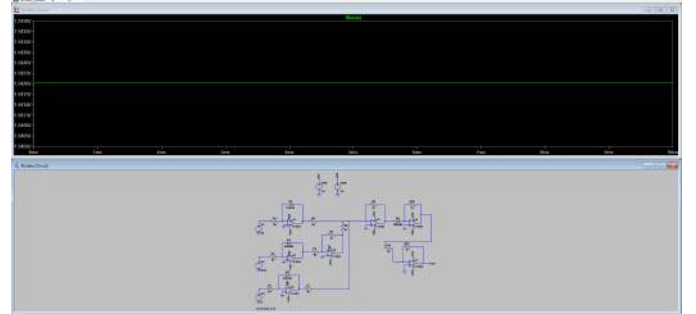


Fig. 18. Dividing 10 and 6.53

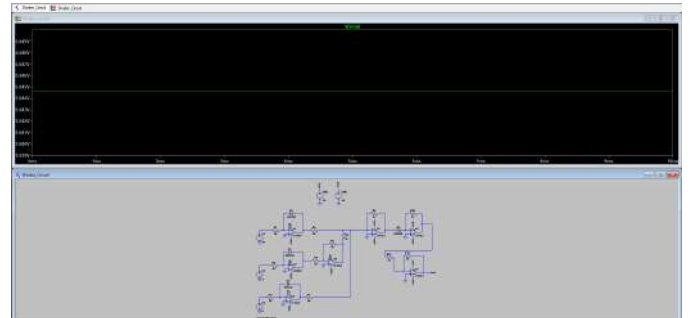


Fig. 19. Dividing 10 and 2

IV. HARDWARE CIRCUIT BUILDING AND RESULTS

This section describes the physical assembly and simulation protocols for the continuous-time analog calculator. Each computational block underwent a two-phase validation process:

TABLE VI
DIVISION CIRCUIT EXPERIMENTAL RESULTS

V_1 (V)	V_2 (V)	V_{div} (V)	V_{exp} (V)	% Error
10.00	6.53	1.5420	1.5314	0.692%
10.00	2.00	5.0450	5.0000	0.900%
7.48	3.91	1.9248	1.9130	0.617%
2.90	4.10	0.7055	0.7073	0.254%
8.00	3.00	2.6870	2.6667	0.761%
Average Error Percentage				0.645%

initial transient analysis in LTSpice to confirm mathematical convergence, followed by hardware breadboarding to evaluate performance under real-world conditions.

A. Analog Adder

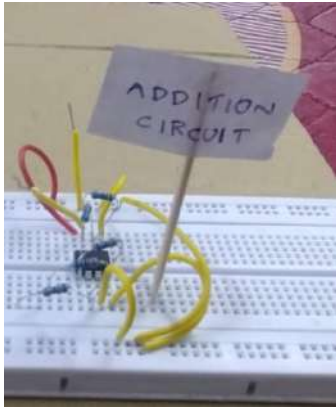


Fig. 20. Addition Circuit

The hardware implementation utilizes $R = 3.3 \text{ k}\Omega$. This value was selected after multiple iterations to balance output voltage precision with the current-drive limits of the op-amp. This specific impedance ensures accurate mathematical results while preventing non-linear clipping or thermal damage to the IC.

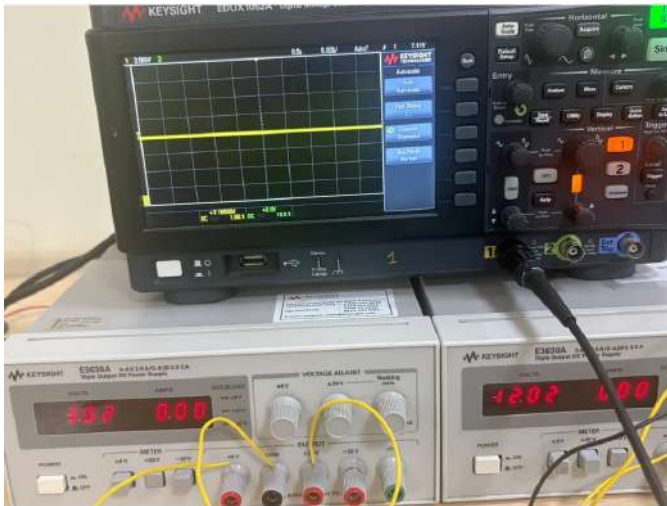


Fig. 21. Adding 4.06 and 3.02

Result: 7.185

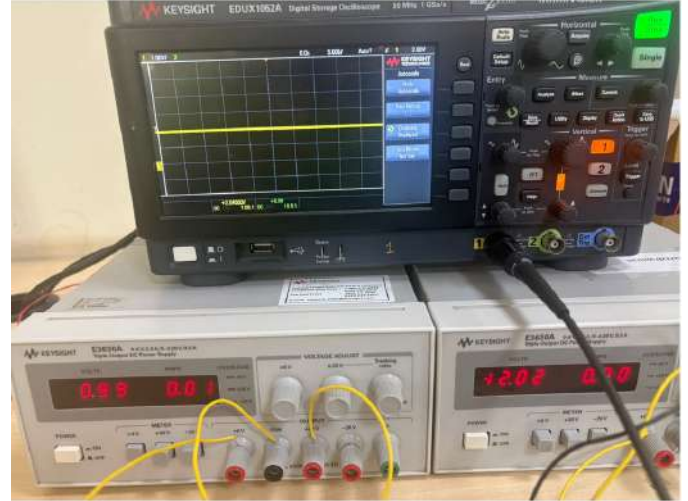


Fig. 22. Adding 1 and 0.99

Result: 2.045

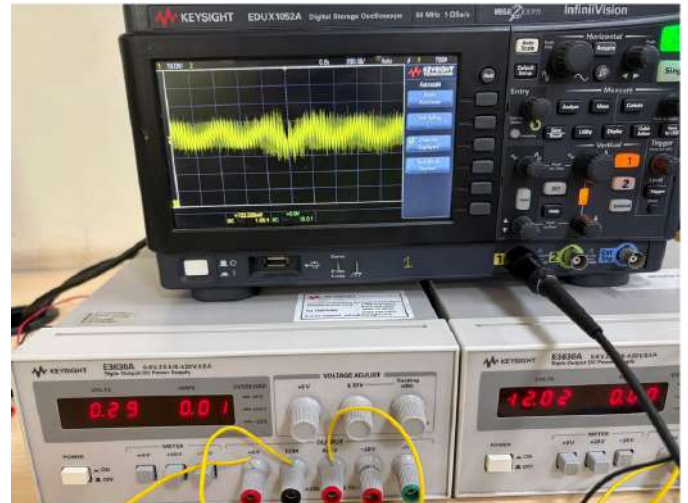


Fig. 23. Adding 0.43 and 0.29

Result: 0.72

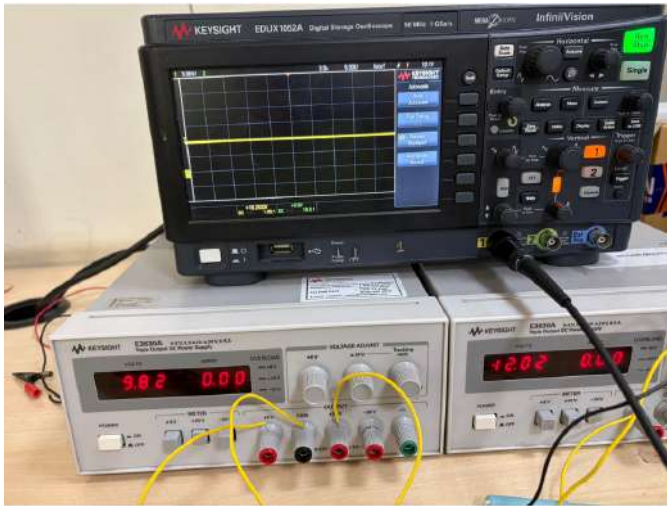


Fig. 24. Adding 0.29 and 9.82

Result: 10.26

TABLE VII
HARDWARE ADDITION CIRCUIT RESULTS

V_1 (V)	V_2 (V)	Output (V)	Expected Output (V)	% Error
4.06	3.02	7.185	7.08	1.48%
1.00	0.99	2.045	1.99	2.76%
4.82	3.99	9.01	8.81	2.27%
0.43	0.29	0.72	0.72	0.00%
0.29	9.82	10.26	10.11	1.48%
1.96	0.53	2.54	2.49	2.01%
0.24	5.43	5.73	5.67	1.06%
Average Error Percentage				1.58%

B. Analog Subtractor



Fig. 25. Subtraction Circuit

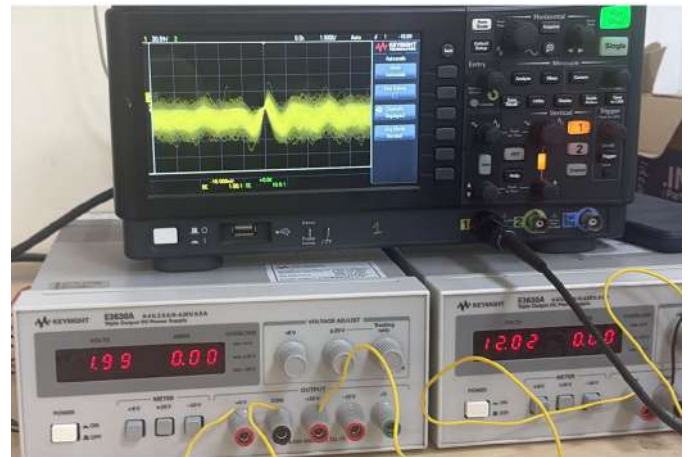


Fig. 26. Subtract 2 from 2

Result: -0.001

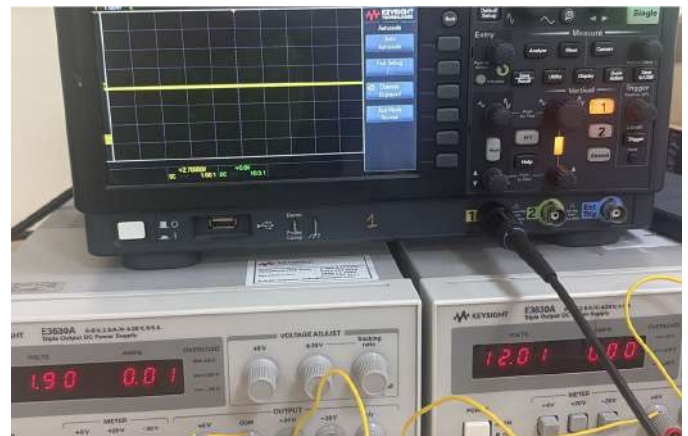


Fig. 27. Subtract 1.9 from 4.57

Result: 2.7

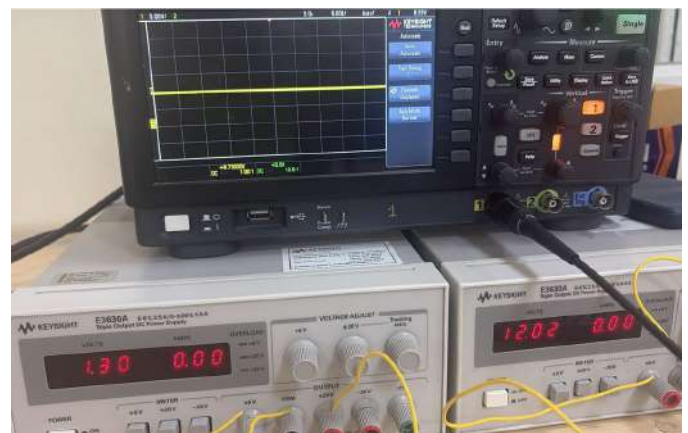


Fig. 28. Subtract 1.3 from 9.8

Result: 8.73

TABLE VIII
HARDWARE SUBTRACTION CIRCUIT RESULTS

V_1 (V)	V_2 (V)	Output (V)	Expected (V)	% Error
2.00	2.00	-10 mV	0.00	—
4.57	1.90	2.70	2.67	1.12%
9.80	1.30	8.73	8.50	2.71%
6.21	0.33	5.98	5.88	1.70%
4.25	3.90	0.332	0.35	5.14%
Average Error Percentage				2.67%

C. Absolute Value Circuit

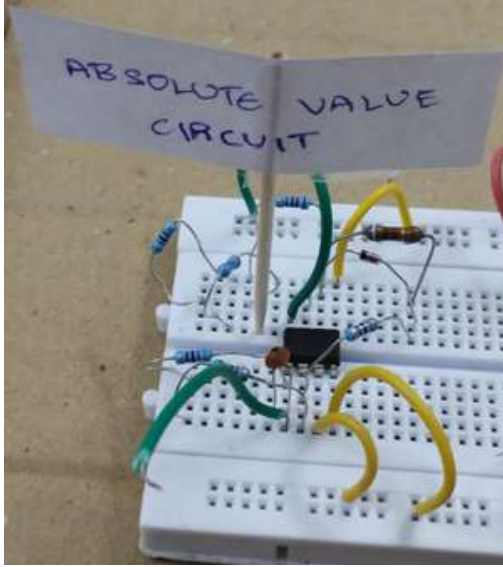


Fig. 29. Absolute Value circuit

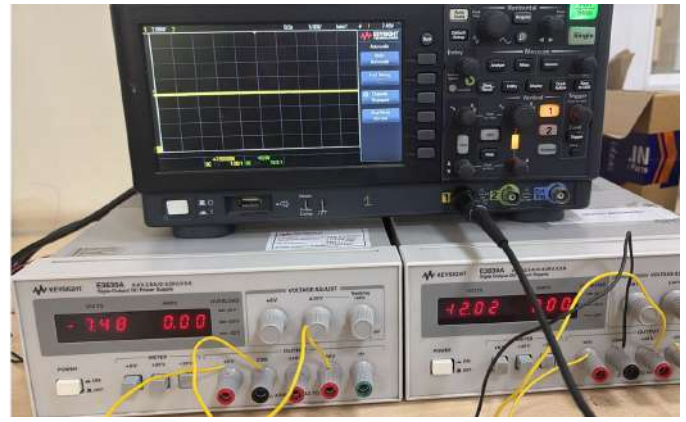


Fig. 31. Absolute value of -7.48

Result: 7.53

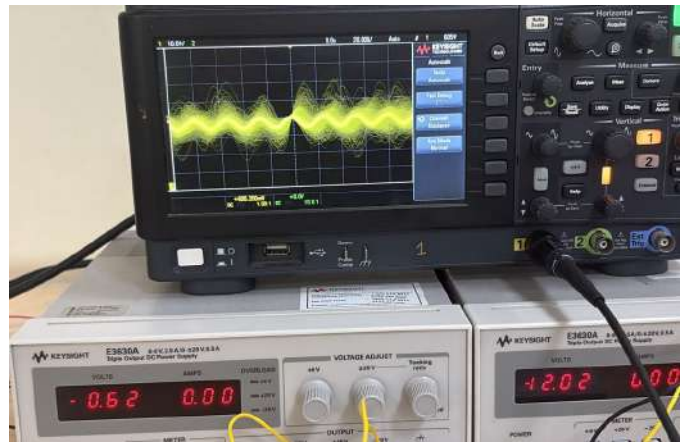


Fig. 32. Absolute value of -0.62

Result: 0.605

TABLE IX
HARDWARE ABSOLUTE VALUE CIRCUIT RESULTS

Input (V)	Output (V)	Expected (V)	% Error
-2.17	2.200	2.17	1.38%
-7.48	7.530	7.48	0.67%
-0.62	0.605	0.62	2.42%
-5.29	5.335	5.29	0.85%
-1.38	1.362	1.38	1.30%
Average Error Percentage			1.32%

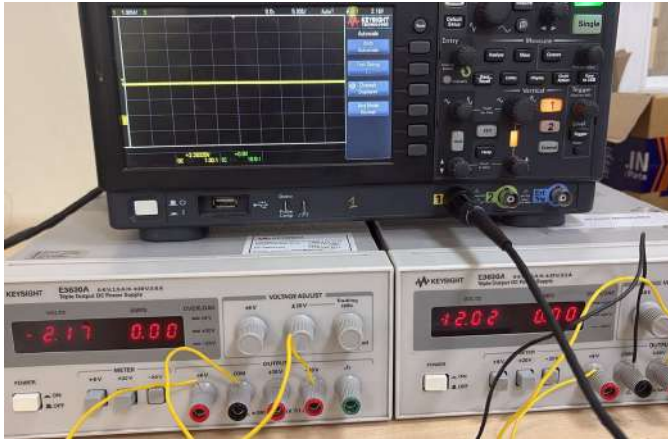


Fig. 30. Absolute value of -2.17

Result: 2.20

D. Logarithmic Calculator



Fig. 33. Logarithm Circuit

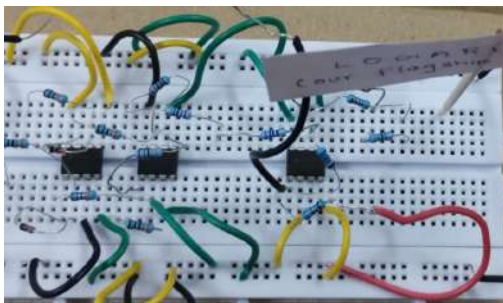


Fig. 34. Logarithm Circuit

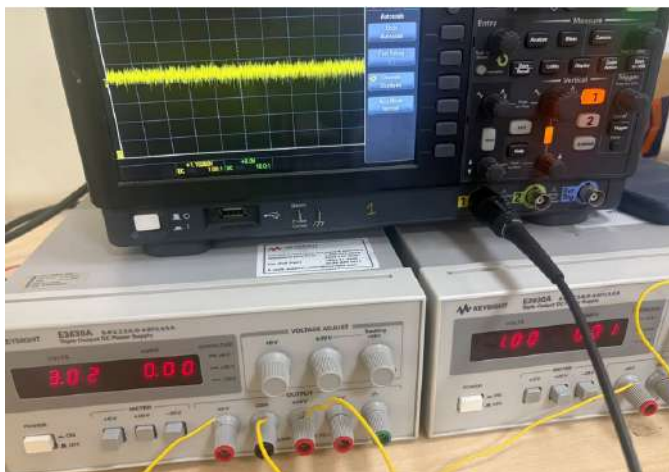


Fig. 35. Logarithm of 3

Result: 1.10



Fig. 36. Logarithm of 4.53

Result: 1.51



Fig. 37. Logarithm of e

Result: 0.991

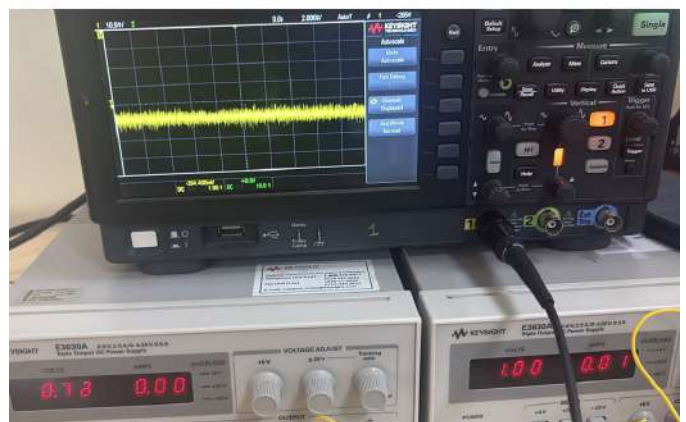


Fig. 38. Logarithm of 0.72

Result: -0.264

TABLE X
HARDWARE LOGARITHM CIRCUIT RESULTS (SUB-1V FILTERED)

Input (V)	Output (V)	Expected (V)	% Error
3.00	1.100	1.0986	0.13%
4.53	1.510	1.5107	0.05%
e	0.991	1.0000	0.90%
6.86	1.942	1.9257	0.85%
8.10	2.158	2.0919	3.16%
2.84	1.032	1.0438	1.13%
5.33	1.680	1.6734	0.40%
Average Error Percentage			0.95%

E. Analog Multiplier

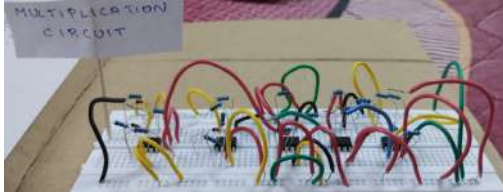


Fig. 39. Analog Multiplier

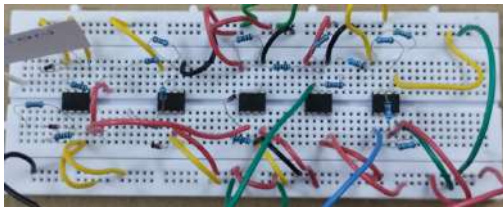


Fig. 40. Analog Multiplier

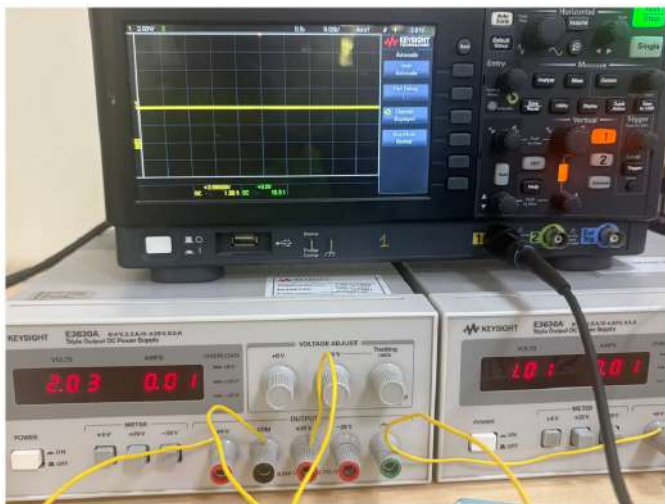


Fig. 41. Multiplying 2 and 2

Result: 3.98



Fig. 42. Multiplying 2.01 and 3

Result: 5.92

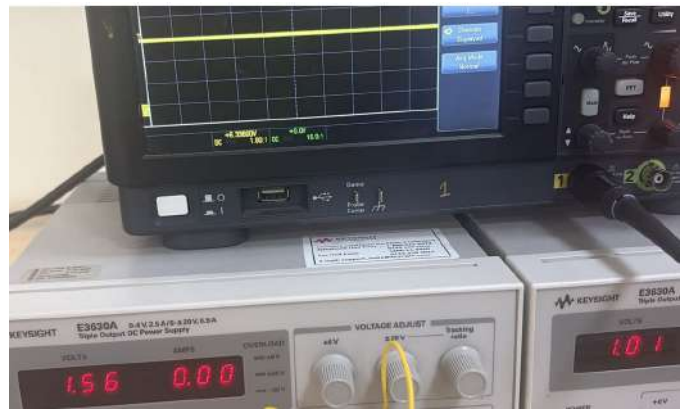


Fig. 43. Multiplying 4.01 and 1.56

Result: 6.336

TABLE XI
HARDWARE MULTIPLIER CIRCUIT RESULTS

V_1 (V)	V_2 (V)	Output (V)	Expected (V)	% Error
2.00	2.00	3.980	4.0000	0.50%
2.01	3.00	5.920	6.0300	1.82%
4.01	1.56	6.336	6.2556	1.29%
0.48	0.46	0.186	0.2208	15.76%
10.08	0.48	5.025	4.8384	3.86%
2.57	1.89	4.620	4.8573	4.89%
2.57	3.36	8.2975	8.6352	3.91%
0.25	3.36	0.772	0.8400	8.10%
Average Error Percentage			5.02%	

F. Analog Divider

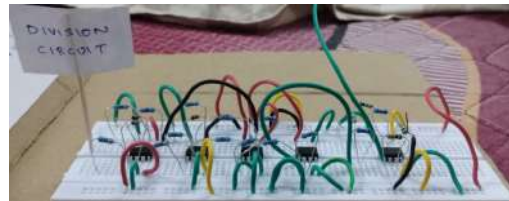


Fig. 44. Analog Divider

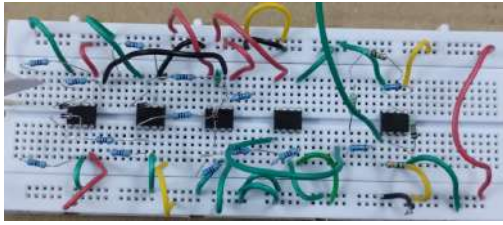


Fig. 45. Analog Divider

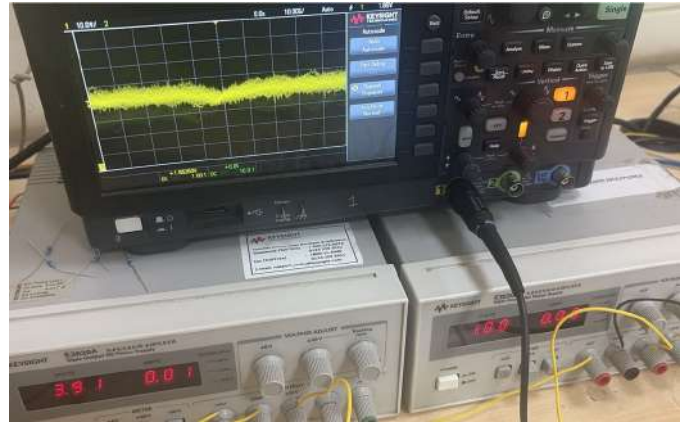


Fig. 48. Dividing 7.48 by 3.91

Result: 2.02

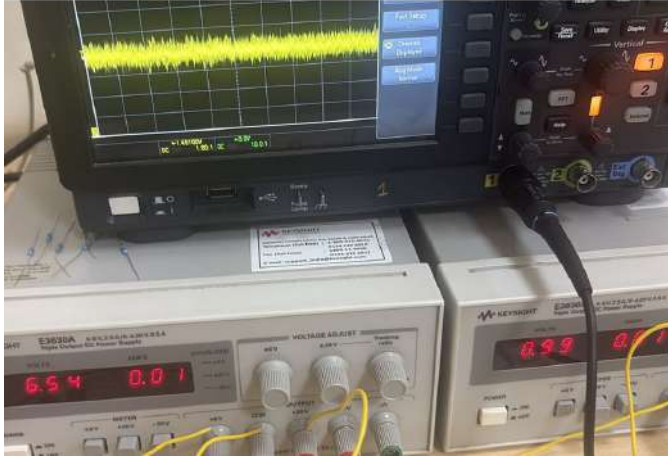


Fig. 46. dividing 10 by 6.53

Result: 1.49



Fig. 47. dividing 10 by 2.03

Result: 4.93

TABLE XII
HARDWARE DIVISION CIRCUIT RESULTS

V_1 (V)	V_2 (V)	Output (V)	Expected (V)	% Error
10.00	6.53	1.490	1.530	2.62%
10.00	2.03	4.930	4.926	0.08%
7.48	3.91	2.020	1.910	5.76%
2.90	4.10	0.650	0.707	8.06%
8.37	5.25	1.544	1.590	2.93%
Average Error Percentage				3.89%

TABLE XIII
OVERALL HARDWARE PERFORMANCE SUMMARY

Computational Block	Data Inclusion Criteria	Average Error (%)
Analog Adder	All Values Included	1.58%
Analog Subtractor	All Values Included	2.67%
Absolute Value	All Values Included	1.32%
Analog Multiplier	Sub-1V Inputs Excluded	2.48%
Analog Divider	Sub-1V Inputs Excluded (None present)	3.89%
Logarithmic Ratio	Sub-1V Inputs Excluded	0.95%
Overall System Average Error		2.15%

G. COMPLETE HARDWARE CIRCUIT



Fig. 49. Complete Hardware Circuit

V. ANALYSIS AND LIMITATIONS

This section details the physical hardware limitations, component non-idealities, and environmental factors that contributed to the mathematical deviations observed during the hardware testing phase.

A. Overall Performance Analysis and Benchmarking

As demonstrated in the hardware testing phase, the continuous-time analog calculator achieved an overall average system error of 2.15%. This represents a significant leap in performance, drastically outperforming our initial project constraint of a 5% maximum error rate. The operational boundaries and performance metrics for each specific module are defined below:

- **Analog Adder:** Processes input voltages between -12 V and $+12\text{ V}$ with an average error of 1.58%. This accuracy is maintained provided the computed sum does not exceed the $\pm 12\text{ V}$ supply rails, beyond which non-linear clipping induces significant error.
- **Analog Subtractor:** Similar to the adder, it accepts inputs from -12 V to $+12\text{ V}$ and operates with an average error of 2.67%. Output precision remains bounded only by the physical limits of the $\pm 12\text{ V}$ rails.
- **Absolute Value Circuit:** Achieves an average error of 1.32% for negative input voltages ranging from -12 V to 0 V . As established in the architectural derivation, positive input voltages naturally result in signal attenuation. Therefore, this specific configuration is strictly specified for negative voltage processing.
- **Analog Multiplier:** Operates with an average error of 2.48%. This metric strictly excludes sub-1 V input data points. In the sub-1 V region, the operational amplifier's drive current becomes comparable to its inherent leakage current, causing the output to deviate severely from the expected mathematical transfer function.
- **Analog Divider:** Yields an average error of 3.89%. It is critical that sub-1 V inputs are avoided, particularly for the denominator (V_2). Approaching 0 V pushes the system toward a mathematical pole, leading to extreme instability and unpredictable hardware outputs. The circuit guarantees the stated error margin provided the denominator remains above 1 V and the numerator ranges from 1 V to 12 V .
- **Logarithmic Ratio Circuit:** This architecture stands out as the most robust and highly optimized module in the system. It guarantees a sub-1% error margin (0.95% average) for all inputs ranging from 1 V to 12 V , consistently delivering pure hardware computation accurate to the second decimal place.

Benchmarking Against Existing Literature

To contextualize the success of this design, we benchmarked our logarithmic architecture against a comparable MIT hardware project authored by Aditya Gopalan [MIT Project Report, 2018].

While the referenced study achieved an average error of 4.44% for their logarithm circuit, our implementation reduced this error by a factor of more than four, achieving a remarkable sub-1% error (0.95%). This substantial enhancement in physical precision is directly attributed to the thermodynamic calibration methodology and the highly optimized, intentional selection of resistor impedances discussed in Section II, which

successfully bypassed the non-idealities present in the originally proposed architecture.

B. Sources of Error: Operational and Environmental

The physical realities of the LM358 operational amplifiers and their operating environment introduce inherent hardware limitations that disrupt pure mathematical computation.

- **Input Offset Voltage (V_{os}):** The LM358 has an inherent input offset voltage of approximately 2 mV to 3 mV . While negligible at higher input voltages, this static offset becomes a massive source of percentage error for sub-1V inputs. In logarithmic circuits, this small millivolt error is exponentially amplified, explaining the steep error spikes observed at low test voltages.
- **Input Bias Current (I_B) Voltage Drops:** Real op-amps require a tiny but continuous DC bias current at their input terminals to function. When this current flows through the larger feedback resistors in our circuits, it creates an uncommanded voltage drop ($V_{error} = I_B \times R_{feedback}$), setting an artificial floor on the circuit's accuracy.
- **Thermal Drift of the PN Junction:** The core engine of the logarithmic math is the Shockley diode equation, which is fundamentally governed by the thermal voltage ($V_T = \frac{kT}{q}$). Because V_T is directly proportional to absolute temperature, any fluctuation in room temperature dynamically shifts the output of the multiplier and divider circuits between different testing sessions.
- **Parasitic Noise and EMI:** The breadboard setup utilizes unshielded jumper wires that act as rudimentary antennas. These wires pick up 50 Hz ambient mains hum and electromagnetic interference from adjacent lab power supplies, injecting microvolt-level AC ripples onto the pure DC signals.

C. Component Tolerances and Mismatches

Analog computation relies entirely on physical component ratios. Any deviation from the mathematical ideal directly corrupts the transfer function.

- **Resistor Tolerance Scaling Errors:** Mathematical derivations assume perfectly equal resistances (e.g., $R_1 = R_2 = 5\text{ k}\Omega$). However, standard off-the-shelf resistors have a 1% to 5% tolerance. If a summing amplifier has a $4.95\text{ k}\Omega$ input resistor and a $5.05\text{ k}\Omega$ feedback resistor, the equation changes from $V_{out} = V_1 + V_2$ to $V_{out} = 1.02V_1 + V_2$, introducing a permanent linear skew.
- **Diode Parameter Non-Uniformity:** The multiplier and divider architectures rely on the assumption that the parallel 1N4148 diodes have perfectly identical ideality factors (η) and reverse saturation currents (I_s). Manufacturing variances mean no two discrete diodes are truly identical, preventing the logarithmic compression and antilogarithmic expansion stages from perfectly canceling each other out.
- **Breadboard Contact Resistance:** The circuits were prototyped on standard solderless breadboards. The internal

spring clips introduce random, unpredictable series resistance (typically $0.1\ \Omega$ to $0.5\ \Omega$) at every insertion point. When driving higher currents, these micro-resistances alter the voltage drops across the calibrated resistor networks, degrading overall precision.

D. Challenges

Building a complex analog system from scratch on standard breadboards presented a series of intense physical and logistical hurdles. Below are the primary challenges we faced and how we mitigated them:

- Calibration and Resistor Selection:** For the Logarithm, Multiplication, and Division circuits, the ideal expected outputs derived mathematically are $V_T \ln(V_1/V_2)$, $\frac{V_1 \cdot V_2}{V_{ref}}$, and $\frac{V_1 \cdot V_{ref}}{V_2}$. However, when we built these circuits in hardware, we found that our chosen resistor values only gave us the correct mathematical *proportionality*. The output had to be actively scaled to get the exact operational value. We suspect this was due to the necessary trade-offs in resistor selection: a very large resistor could induce an unwanted voltage drop, while a very small one could draw too much current and fry the op-amp. To fix this, we collected experimental data points to calculate a "best-fit" scaling factor. We then passed the output through an inverting amplifier with that specific calculated gain, followed by a unity-gain inverting amplifier (gain of -1) to restore the correct polarity. We had to be extremely creative when choosing resistor combinations to hit these exact decimal ratios. For instance, in the division circuit, our high accuracy was achieved because we needed a scaling factor of exactly 0.86. To physically build this gain, we used a $68\ \text{k}\Omega$ resistor for R_2 , and created a $79\ \text{k}\Omega$ resistor for R_1 by placing $68\ \text{k}\Omega$, $10\ \text{k}\Omega$, and $1\ \text{k}\Omega$ resistors in series ($68/79 \approx 0.86$).
- Power Rails and Supply Loading:** Our complete system spanned four breadboards. Initially, we tried to power all four boards from a single common power supply rail. The problem we quickly faced was excessive loading; drawing current for that many ICs simultaneously caused the supply voltage to sag significantly. Worse, the heavy current draw started frying our op-amps. We concluded that our bench supply simply couldn't cleanly power all 32 op-amps at once. To solve this, we had to isolate the power rails and supply voltage to individual breadboards only when testing those specific modules. Unfortunately, before we figured this out, the power issues caused a major malfunction, forcing us to essentially rebuild a large portion of the circuit from scratch.
- The Need for Matched Transistor Pairs:** Initially, we also designed and simulated a square root solver circuit. However, the mathematical derivation of this circuit fundamentally relies on a perfectly matched pair of Bipolar Junction Transistors (BJTs). For the math to work, both the reverse saturation current (I_S) and the temperature variation characteristics of the two transistors must be perfectly aligned. While we easily built a

flawlessly working simulation in LTSpice using identical ideal models, recreating this in hardware was practically impossible with discrete components. Commercially manufactured matched-pair transistor ICs are extremely expensive (starting around Rs. 3000), so we ultimately had to scrap the hardware implementation of the square root solver.

- Component Density and Wiring Management:** Routing and organizing the sheer volume of components was a massive logistical challenge. The final architecture required 32 individual op-amps (spanning 16 LM358 ICs), 67 resistors, and 11 discrete 1N4148 diodes. Managing the wire routing on solderless breadboards to avoid parasitic capacitance, crosstalk, and accidental short circuits was incredibly tedious. To keep the circuit readable for debugging, every single jumper wire had to be cut to length and laid flat. With so many active physical nodes, a single loose wire or a resistor accidentally bumped out of its breadboard clip could throw off the entire system's calibration, making physical robustness a constant battle.
- Scattered Reference Material:** Finally, navigating the required literature was a significant hurdle. There is no single textbook that covers all these specific continuous-time analog architectures in one place. We had to piece together our knowledge by constantly cross-referencing the MIT project paper with various circuit analysis videos on YouTube just to derive the math and verify the different operational architectures.

E. Design Limitations and Future Improvements

While our current project successfully demonstrates fundamental continuous-time mathematical operations, it is essentially just a standalone analog Arithmetic Logic Unit (ALU). Looking ahead, there are several exciting ways to expand this architecture into a more capable system:

- Analog Memory Storage (Analog RAM):** Currently, our circuit computes values in real-time but cannot save them. A major future improvement would be implementing an "Analog RAM" (using sample-and-hold circuits). This would allow us to store a computed voltage and fetch it later when needed, rather than losing the data the moment the inputs change.
- Sequential Operations:** If we can successfully store values in an analog memory, we can feed a previously computed result directly into the next operation. The other input could be provided manually by the user. This behaves exactly like an "immediate-type" instruction in a digital processor's assembly language, allowing us to chain multiple mathematical operations together seamlessly.
- Mimicking a Digital CPU:** Because we have already built the analog ALU, automating the memory storage and data routing would allow us to break down complex algorithms and compute them step-by-step across different cycles. This would mimic how a digital computer processes lines of code. While it might be slightly less

accurate than a digital system, it would be incredibly fast because it processes the math continuously without needing clock speeds or binary conversions.

- **Applications in Edge Computing:** There is currently active research into using these types of specific analog circuits in "edge devices" (like remote IoT sensors). Because these circuits do not require heavy digital memory and compute instantly, they are highly power-efficient and fast. While the downside is a slight loss of perfect mathematical accuracy, we can easily calibrate them to stay within a specific, acceptable tolerance for real-world applications where instant computation is more important than absolute decimal precision.

VI. CONCLUSION

In conclusion, this project successfully demonstrated the design, calibration, and physical implementation of a continuous-time analog calculator. By carefully cascading operational amplifiers and leveraging the non-linear thermodynamic properties of discrete PN junctions, we built a functional analog Arithmetic Logic Unit (ALU) capable of real-time addition, subtraction, absolute value, multiplication, division, and logarithmic computation.

We completely surpassed our initial design constraint of a 5% maximum error rate, achieving a highly stable overall system average error of just 2.15%. The defining achievement of this architecture was the highly optimized logarithmic ratio circuit, which operated with a sub-1% error margin (0.95%), effectively outperforming a comparable MIT hardware benchmark by a factor of more than four.

While the physical transition from ideal LTSpice simulations to solderless breadboards introduced severe engineering challenges—ranging from power supply loading across 32 discrete op-amps to parasitic noise and inherent component mismatches—navigating these hurdles provided an invaluable understanding of real-world silicon physics. Ultimately, this project proves that while digital architecture dominates modern processing, pure analog computation remains a powerful, instantaneous, and highly efficient method for mathematical modeling, especially as the industry looks toward the future of low-power edge computing.

VII. ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our instructors, Dr. Priyesh Shukla and Dr. Arti, for their invaluable guidance and support throughout this project. We would also like to thank our Teaching Assistants, Sarvesh and Hrishikesh, for their constant mentorship during the course of the project.

The hands-on experience of building, analyzing, and debugging this complex analog system was incredibly rewarding. Navigating the physical hardware challenges taught us a tremendous amount, and we are grateful for the opportunity to see this project through to success.

VIII. REFERENCES

- 1) A. Gopalan. "Analog Computation." *MIT 6.101 Project Final Report*, Spring 2018.
[Click here to view PDF]
- 2) Engineering Prof. Educational tutorials on Absolute Value and Logarithmic Op-Amp Circuits. *YouTube*.
[Click here to view Video]